

Lecture 10 – Model Selection and Regularization

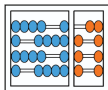
[ISLP 6.1, 6.2]

MC886 – Machine Learning

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Today's summary

Subset selection

Choosing the optimal model

Regularization (shrinkage) methods: Ridge regression

Regularization (shrinkage) methods: The lasso

Alternative fitting procedures for linear regression

- ❖ In the regression setting, the standard linear model:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon \quad (6.1)$$

is typically adjusted using least squares

- ❖ However, one might want to use another fitting procedure, for the sake of:
 - **Model interpretability**
(can be attained through removal of irrelevant variables)
 - **Prediction accuracy**
(can be improved by putting brakes into the fitting procedure)

Three classes of methods

1. Subset selection

- Train a model only with a subset of variables (e.g. forward selection)

2. Regularization

- Shrinkage of coefficients towards zero (e.g. ridge regression)

3. Dimension reduction

- Projection of p variables into a M -dimensional space, $M < p$ (e.g. PCA - *we'll study it latter in this course*)

Variable subset selection (“feature selection”)

Select a subset of the p variables. For example, consider the following dataset:

$$\left\{ \left(\begin{pmatrix} x_{11} \\ x_{12} \\ x_{13} \end{pmatrix}, y_1 \right), \left(\begin{pmatrix} x_{21} \\ x_{22} \\ x_{23} \end{pmatrix}, y_2 \right), \left(\begin{pmatrix} x_{31} \\ x_{32} \\ x_{33} \end{pmatrix}, y_3 \right) \right\}$$

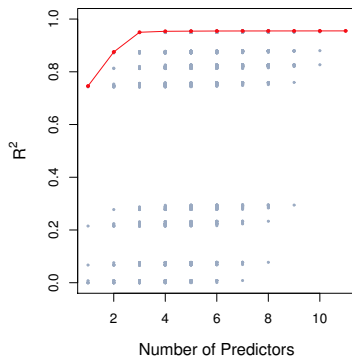
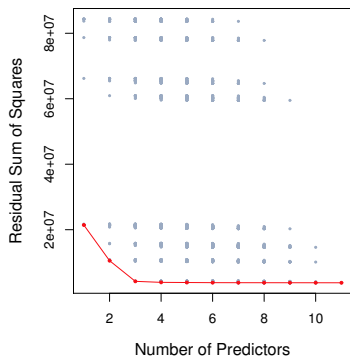
If we select variables 1 and 3, then we train the model with:

$$\left\{ \left(\begin{pmatrix} x_{11} \\ x_{13} \end{pmatrix}, y_1 \right), \left(\begin{pmatrix} x_{21} \\ x_{23} \end{pmatrix}, y_2 \right), \left(\begin{pmatrix} x_{31} \\ x_{33} \end{pmatrix}, y_3 \right) \right\}$$

Algorithm 6.1: Best subset selection

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \dots, p$:
 - a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
 - b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here best is defined as having the smallest RSS, or equivalently largest R^2 .
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using an optimality criterion.

Best subset selection on Credit dataset



- ❖ Each dot is the given error for a given subset of variables (subset size is given by the x axis)
- ❖ Dots in red are the best subset for a given subset size

Finding optimal subset for large p has an issue:

Time complexity (*why?*)

❖ **Answer:** *The number of possibilities is 2^p
(exponential on # variables)*

■ 2^{240} is larger than # protons in the universe!

❖ **Solution:** *Constraint on the number of evaluated subsets*

Algorithm 6.2: Forward stepwise selection

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors.
2. For $k = 0, 1, \dots, p - 1$:
 - a) Consider all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
 - b) Choose the best among these $p - k$ models, and call it $\mathcal{M}_{k+1}$.
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using an optimality criterion.

Forward stepwise selection

- ❖ Before the first iteration, it fits the null model
- ❖ At the k th iteration, it fits $p - k$ models. This amounts to a total of:

$$\begin{aligned} 1 + \sum_{k=0}^{p-1} (p - k) &= 1 + p(p + 1)/2 \\ &= O(p^2) \end{aligned}$$

- ❖ A big improvement on the $O(2^p)$ of the best subset approach!

First four models for best and forward stepwise selections

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income, student, limit	rating, income, student, limit

Algorithm 6.3: Backward stepwise selection

1. Let $\mathcal{M}_p$ denote the *full model*, which contains all p predictors.
2. For $k = p, p - 1, \dots, 1$:
 - a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k - 1$ predictors.
 - b) Choose the best among these k models, and call it $\mathcal{M}_{k-1}$. Here best is defined as having smallest RSS or highest R^2 .
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using an optimality criterion.

Backward stepwise selection

- ❖ As the forward stepwise selection, it fits $O(p^2)$ models
- ❖ However, it requires that $n > p$ (to allow fitting $\mathcal{M}_p$)
- ❖ Backward and forward selections can be combined into hybrid approaches (try to avoid the greedy limitation of them)
- ❖ Now, what would be an “optimality criterion”???
(Algorithms 6.1–6.3)

Choosing the optimal model

- ❖ RSS and R^2 on training set are not suitable for evaluating the best model
- ❖ Therefore, we need to look to the test error, either:
 1. *Indirectly*, through an adjustment on the training error
 2. *Directly*, using a validation set or cross validation (CV)
- ❖ We'll see both of them now

Adjustment of the training error

We'll study four of those approaches:

1. Mallow's C_p
2. Akaike information criterion (AIC)
3. Bayesian information criterion (BIC)
4. Adjusted R^2

Mallow's C_p

- ❖ For an adjusted least squares model with d predictors, the C_p estimate of the test MSE is given by:

$$C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2), \quad (6.2)$$

where $\hat{\sigma}^2$ is an estimate of the variance error ϵ in Eq. 6.1

- ❖ We estimate $\hat{\sigma}$ using the residual standard error (RSE):

$$\hat{\sigma} = RSE = \sqrt{RSS/(n-2)}$$

- ❖ We include a penalty term $2d\hat{\sigma}^2$ which is a function of the number of used variables

Akaike Information Criterion (AIC)

- ❖ AIC is defined for a large class of models fit by maximum likelihood (e.g. logistic regression). It is defined as:

$$AIC = -2 \log L + 2d,$$

where L is the maximized value of the likelihood function

- ❖ *Special case:*

In the linear model (Eq. 6.1) with Gaussian errors, maximum likelihood and minimization of MSE are the same thing

- In this special case, AIC is equivalent to C_p

Bayesian information criterion (BIC)

- ❖ Derived from the Bayesian point of view, it is defined as:

$$BIC = \frac{1}{n}(RSS + \log(n)d\hat{\sigma}^2) \quad (6.3)$$

- ❖ Similar to C_p , replacing the penalty term constant ($\log(n)$ in the place of 2)
- ❖ This means BIC penalizes heavier subsets of variables where $n > 7$, since the latter implies $\log(n) > 2$

Adjusted R^2

- ❖ Usually, R^2 is defined as:

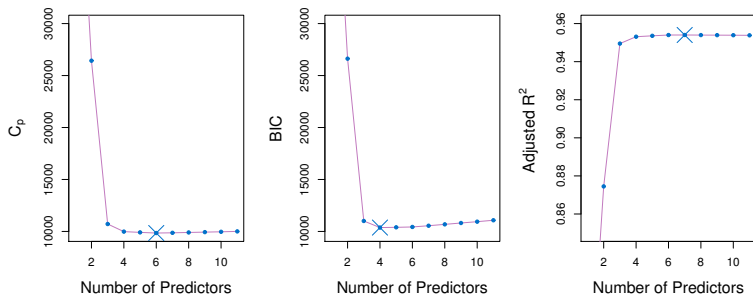
$$R^2 = 1 - RSS/TSS,$$

where RSS is the residual sum of squares ($\sum_{i=1}^n (y_i - \hat{y})^2$) and TSS is the total sum of squares ($\sum_{i=1}^n (y_i - \bar{y})^2$)

- ❖ The *adjusted* R^2 is computed as:

$$\text{Adjusted } R^2 = 1 - \frac{RSS/(n - d - 1)}{TSS/(n - 1)} \quad (6.4)$$

C_p vs. BIC vs. adjusted R^2 (Credit dataset)

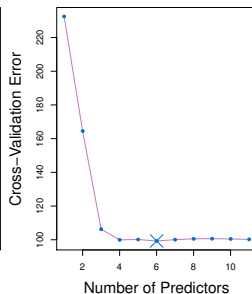
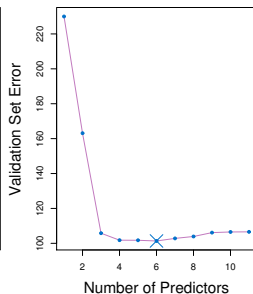
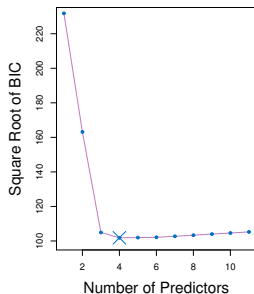


Remark the effect of a heavier penalization in BIC
(in comparison to the one applied by C_p) for $d > 4$!!!

Validation set, cross validation (CV)

- ❖ Direct estimate of the test error
- ❖ Fewer assumptions than C_p , AIC, BIC and adjusted R^2
- ❖ For CV, Algorithms 6.1–6.3, we run step 3 using either the validation set approach, leave-one-out CV (LOOCV) or K -fold CV for choosing the best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$
- ❖ The ISLP book has an alternative, somewhat more complicated, approach

BIC vs. validation set approach vs. CV (Credit dataset)



Regularization (shrinkage) methods

- ❖ Using subset selection, use adjusted a linear model using a subset containing d of the p predictors
- ❖ Here, we'll avoid overfitting by putting constraints (i.e. shrinking) the coefficients
- ❖ We'll see two classic regularization (shrinkage) techniques:
 - *Rigde regression*
 - *Least absolute shrinkage and selection operator (lasso)*

Ridge regression

- ❖ Least squares fitting produces estimates for $\beta_0, \beta_1, \dots, \beta_p$ using values that minimize:

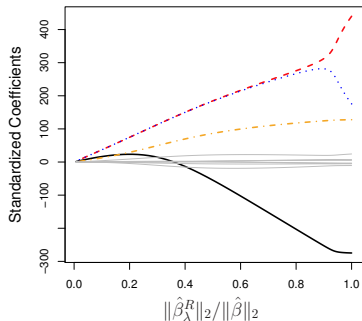
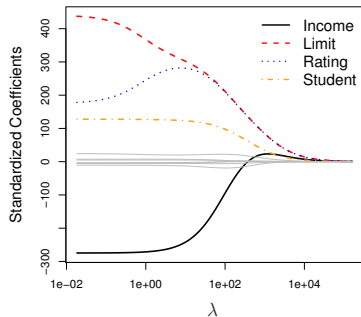
$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

- ❖ Ridge regression estimates $\hat{\beta}^R$ that minimize:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad (6.5)$$

where $\lambda \geq 0$ is a *tuning parameter*

Standardized ridge regression on Credit data

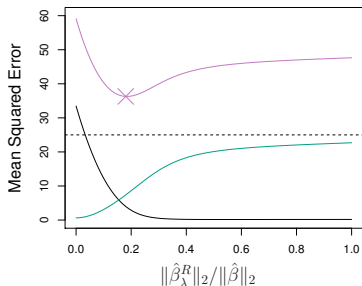
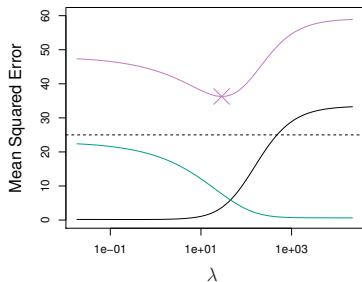


$\|\beta\|_2 = \sqrt{\sum_{j=1}^p \beta_j^2}$ is the ℓ_2 norm
(it measures the distance of β from zero)

Why is ridge regression better than least squares?

- ❖ Short answer: it tackles better the *bias-variance trade-off!!!*
- ❖ The tuning parameter λ controls the trade-off
- ❖ Let's inspect a graphical example

Ridge regression on a simulated dataset



Black curve: squared bias

Green curve: variance

Purple curve: test MSE

Dashed line: minimum possible MSE

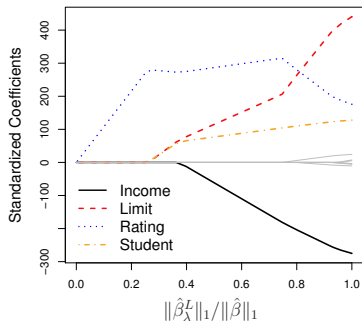
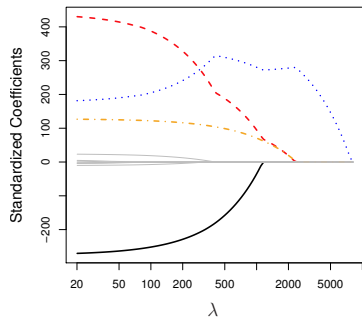
The lasso

- ❖ Con of ridge regression: it doesn't do subset selection!
- ❖ It can be an interpretability challenge for a large p
- ❖ An alternative, called *lasso*, is formulated as:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|, \quad (6.5)$$

where $\lambda \geq 0$ is a *tuning parameter*

Standardized lasso coefficients on the Credit data

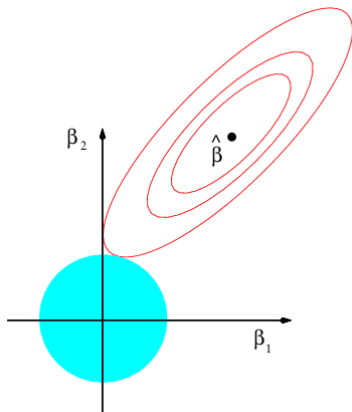
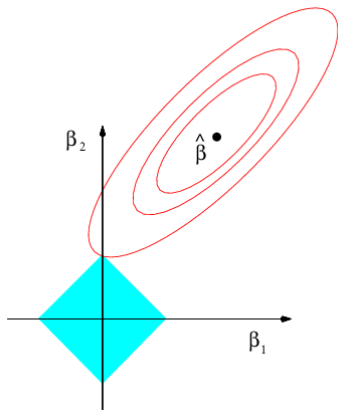


$|\beta|_2 = \sum_{j=1}^p |\beta_j|$ is the ℓ_1 norm
(it measures the distance of β from zero)

Lasso

vs.

ridge regression



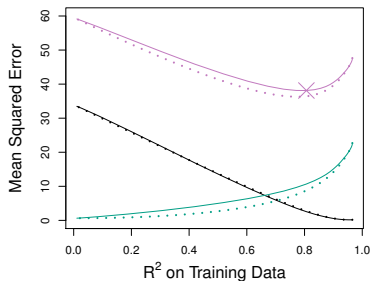
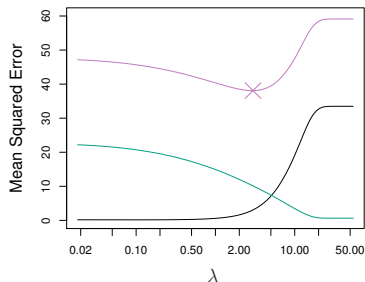
Countour ellipses are RSS, solid blue areas are constraint regions:

$$|\beta_1| + |\beta_2| \leq s \text{ (lasso)}$$

$$\beta_1^2 + \beta_2^2 \leq s \text{ (ridge regression)}$$

Lasso vs. ridge regression

(simulated data; they have 40 variables, all of them related to response)



Solid and dashed lines are lasso and ridge regression, respectively

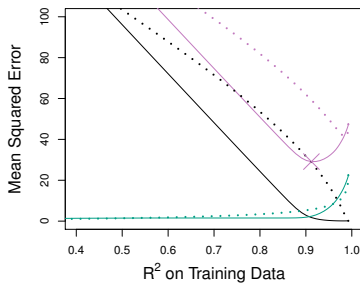
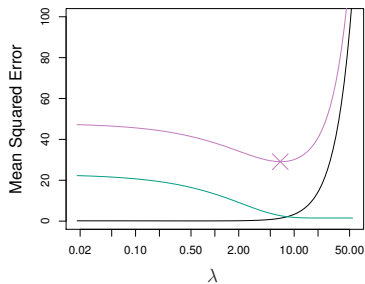
Black curve: squared bias

Green curve: variance

Purple curve: test MSE

Lasso vs. ridge regression

(same as previous graph, but with only 2 variables related to response)



Solid and dashed lines are lasso and ridge regression, respectively

Black curve: squared bias

Green curve: variance

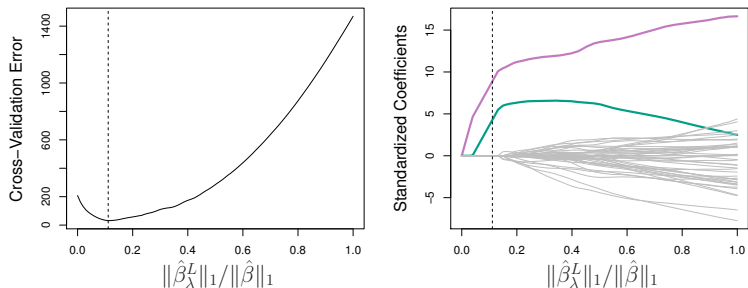
Purple curve: test MSE

Selecting the tuning parameter

- ❖ λ is a *hyperparameter* that must be tuned
- ❖ Common approach: to establish a grid of λ values and carry out cross validation on each value of λ
- ❖ We select the λ value with lowest value and refit the model with all training data and that λ value
- ❖ Some hyperparameter search tools:
 - GridSearchCV (Scikit-learn)
 - Optuna (open source hyperparameter optimization framework)

10-fold CV MSE for lasso

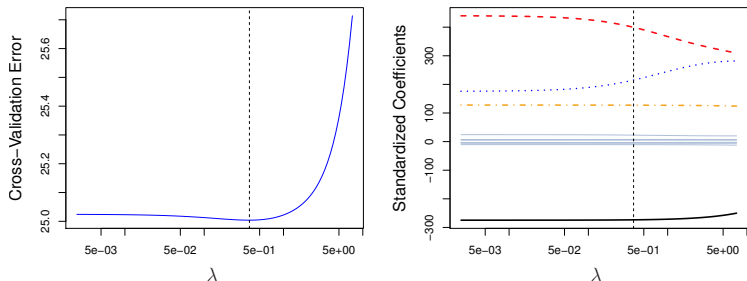
(same data as previous graph, with only 2 variables related to response)



Signal variables in color, noise variables in gray

Dashed line indicates the lasso fit for which CV error is the smallest

CV for ridge regression on Credit data



Signal variables in color, noise variables in gray

Dashed line indicates the lasso fit for which CV error is the smallest